

A CONTINUOUS GRAPHIC DISPLAY OF SEISMIC MIGRATION

Introduction

For many years FPS array processors have been used successfully for seismic processing. They have reached a mature level of usage, with a large number of processing centers using FPS APs 24 hours per day, running highly optimized programs. Shown here is an efficient implementation of one such standard procedure, Kirchhoff-Summation Migration. This article describes the software process during migration, and describes the system configuration used to demonstrate this technique.

Seismic Migration

As seismic data is collected, it is organized as a two-dimensional array, where one dimension represents the variations of amplitude over time, and the second dimension represents the location on the surface of the earth (linear position). To derive from this data an understanding of the sub-surface structure, seismic migration is used to create a reflection event map in spatial coordinates, repositioning each data point (or reflective value) correctly on the seismic section. Accurate migration requires both high-quality velocity estimates and an algorithm that can properly apply the velocity information to unmigrated data. The process of developing the velocity model usually involves repeated migrations, refining the model at each step.

With the availability of high-performance graphic displays and the intelligent I/O co-processor, the migration process can be demonstrated in progress. This is an invaluable tool to software developers, because it provides immediate presentation of the effects of various experimental algorithms. In addition, it is a valuable training system, providing a visual reference to the understanding of migration.

Kirchoff-Summation Migration

Most seismic migration processes are derived from the "wave equation" description of seismic wave propagation. The Kirchhoff diffraction equation is an expression of reflected wave amplitude as distance from a diffracting aperture, and media velocity (Sheriff, 1973). French

(1975) showed that, with some refinement, the heuristic migration approach of summing amplitudes along diffraction hyperbolas is equivalent to the Kirchhoff-integral solution of the wave equation. The migration algorithm used here consists of integrating a symmetrical area around each trace, with a hyperbolic distribution area. This special mapping function represents the diffracting wave front from the Kirchhoff diffraction equation. This technique has been successfully applied to both two- and three-dimensional migration (Gibson, Larner and Levin, 1979).

In two dimensions, Kirchhoff-Summation migration is accomplished (in part) by summing trace amplitudes along hyperbolic diffraction paths and placing each sum at the apex or minimum time position of the associated summation path. Complete implementations contain weighting functions and filtering prior to summation. In three dimensions, the summation process involves amplitudes at positions on hyperboloidal diffraction surfaces. Each sum is then placed at the apex of the hyperboloid sheet. Throughout the migration process, the accuracy of the velocity model is critical to ensure the correct placement of the hyperbolic diffraction pattern; otherwise, misleading distortion is introduced.

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Implementation

The seismic data is stored on disk files in trace-sequential format. The migration process is run on the FPS-5205, using a series of routines written in VFC (for compatibility with the FPS-100), FPS Math Library, and APAL (see Figure 1). The inter-process buffers are sent to the display system, while the output migrated data also can be sent to a disk file for later use. The program is designed to maximize throughput in a batch environment. The graphic display software is executed on the GPIOP, and hence has little impact on system throughput. The GPIOP is synchronized with the AP to fix, pack, and place the data onto the display memory. The velocity interpolation and Kirchhoff-Summation migration hyperbolic mapping contain routines coded in APAL; the graphics executive is written in GPAL.

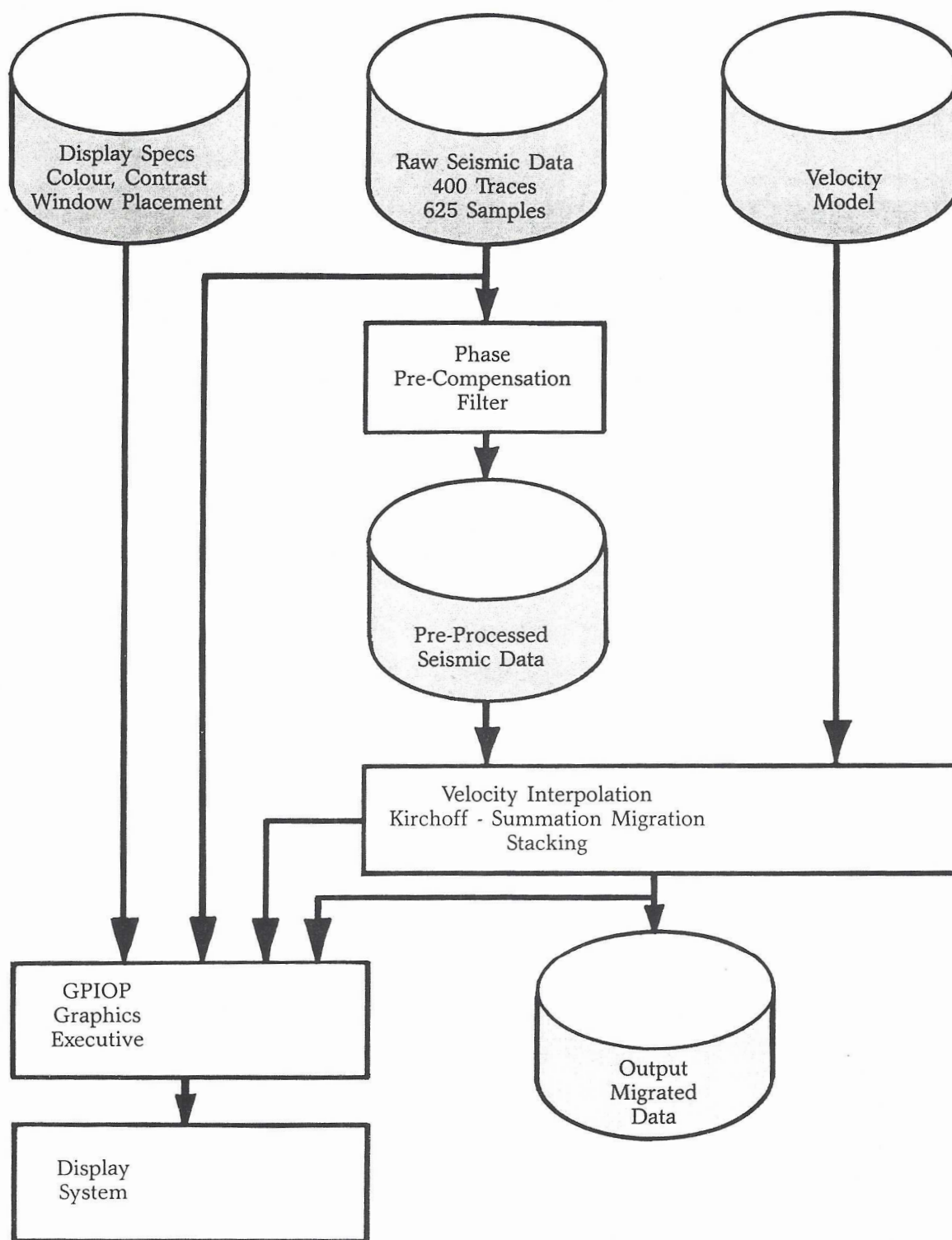


Figure 1. Data Flow Diagram

Display Format

In the example shown in Figure 2, the input data is displayed in the left window, and intermediate process data in the right window. To display as much of the seismic section as possible, a variable density plot format is used. As output traces are computed, they are sent back into the left window, overwriting the unmigrated data, so the step-by-step correspondence can be seen. All of the display data is color-coded according to the buffer the data came from, so that it can be easily identified. The current migration center trace is highlighted in bright red for further clarity. For each migrated output

trace sent to the left window, one complete hyperbolic summation pass is shown on the right window. Because all data shown in the right window is summed for output, the location of contributing events can be seen as they are computed. Since the lateral variation data is small from one trace to the next, the right frame is cleared before each pass to clearly show the diffraction pattern.

The seismic data shown in Figure 2 is 200 traces of 8ms per point data, with the top 325 samples of each trace displayed. The migration is over 400 input traces, because the summation process uses data within a 100-trace window on either side of the center trace.

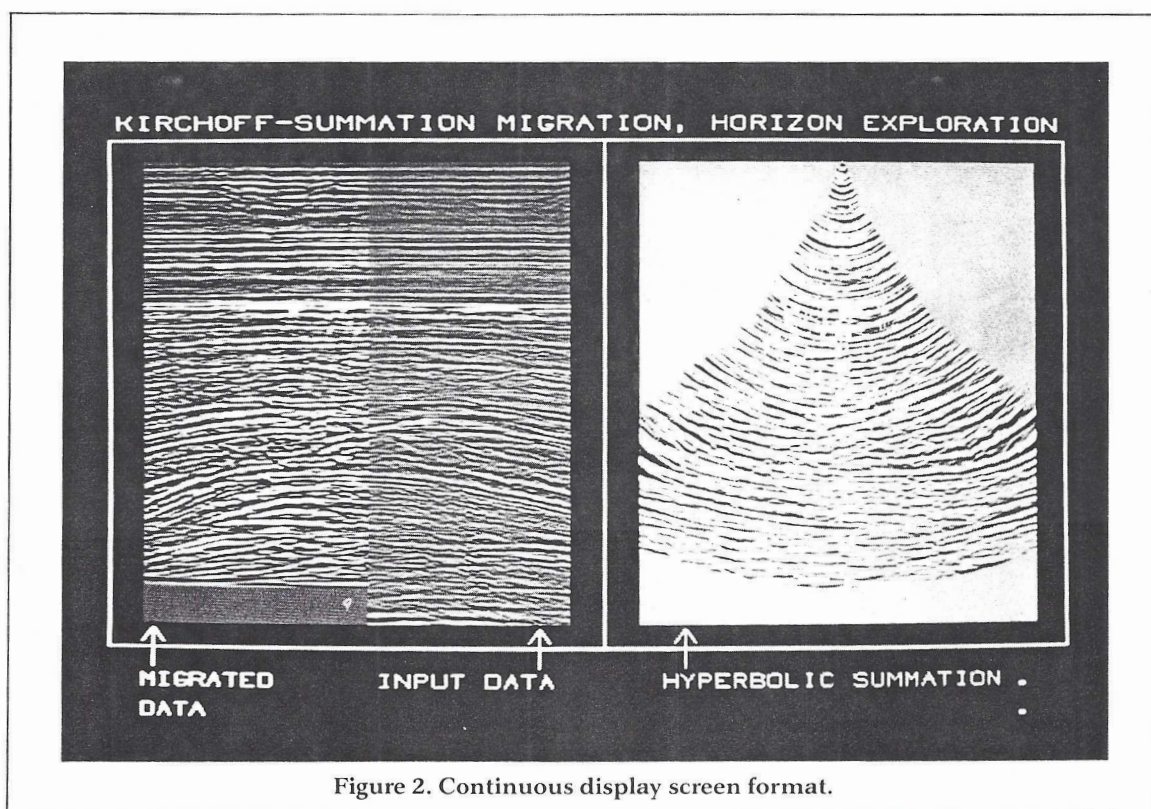


Figure 2. Continuous display screen format.

Equipment Configuration

The computer system used for this project is a general purpose super-minicomputer with an AP and display system (see Figure 3). The host computer is a VAX 11/780 with RA-80 disk drive and streaming tape. The

FPS-5205 performs the primary role of seismic migration. A KOALA frame store (from Bear Systems Ltd.) is directly controlled by the GPIOP, using a high-speed interface to the 1.5 Mbyte pixel memory. The KOALA generates a high-resolution, full-color video image for the display monitor.

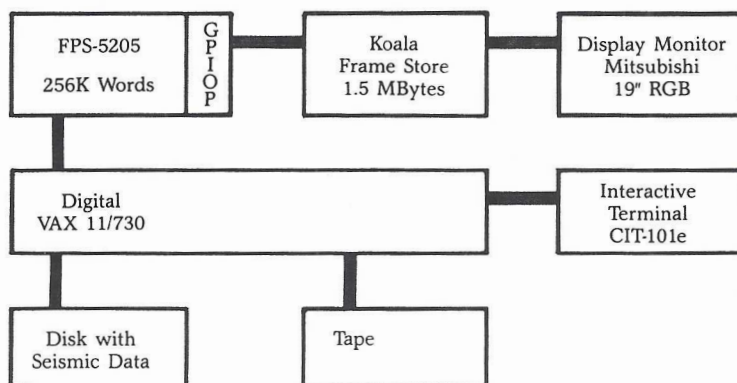


Figure 3. System Configuration

Summary

The software and hardware used in this system is available for demonstration of a graphic display of the migration process. The instant display of process data buffers allows rapid debugging and refinement of new algorithms. In addition, this system serves to educate and explain a technique that is otherwise difficult to visualize. This article has described one practical use of a very flexible system for the continuous display of a seismic migration process.

Acknowledgments

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